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Course code: CSA0925

Course Name: Programming in JAVA

CO1 ASSESSMENT TOOL-3

Coding Challenge

Question-1:

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

Jayaraman Renuka Devi
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CO1-AT3-Coding Challenge

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QUESTIONS

Q1 ✓
Smart University Payroll Management System
25 marks

Q2 ✓
Smart Vehicle Insurance Claim System
25 marks

2/2 answered

Q1 Smart University Payroll Management System 25 marks

A university employs three categories of staff: Teaching Faculty, Technical Staff, and Visiting Faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a superclass Employee with the following private data members: employeeId employeeName department basicSalary Provide: Default constructor Parameterized constructor Getter and Setter methods displayEmployee() calculateSalary() Create the following subclasses: TeachingFaculty teachingHours incentivePerHour Salary: basicSalary + (teachingHours × incentivePerHour) TechnicalStaff overtimeHours overtimeRate Salary: basicSalary + (overtimeHours × overtimeRate) VisitingFaculty lecturesHandled paymentPerLecture Salary: lecturesHandled × paymentPerLecture Override calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display: Employee details Monthly salary Highest-paid employee Average salary Search an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

CSE

50000

20

Your Code

```

1 import java.util.*;
2
3 class Main
4 {
5     public static String readInput(Scanner sc)
6     {
7         String s = "";
8
9         while(sc.hasNextLine())
10        {
11            s = sc.nextLine().trim();
12
13            if(!s.isEmpty())
14            {
15                return s;
16            }
17        }
18
19        return s;
20    }
21
22    public static void main(String args[])
23    {
24        Scanner sc = new Scanner(System.in);
25
26        int n = Integer.parseInt(readInput(sc));
27

```

Input

```

3
TeachingFaculty
101

```

Output

```

Employee Details
Employee ID : 101
Name : Dr.Arun
Department : CSE
Monthly Salary : 60000.0

```

```

238
239 class VisitingFaculty extends Employee
240 {
241     private int lecturesHandled;
242     private double paymentPerLecture;
243
244     public VisitingFaculty()
245     {
246
247     }
248
249     public VisitingFaculty(int employeeId,String employeeName,
250                             String department,double basicSalary,
251                             int lecturesHandled,double paymentPerLecture)
252     {
253         super(employeeId,employeeName,department,basicSalary);
254         this.lecturesHandled = lecturesHandled;
255         this.paymentPerLecture = paymentPerLecture;
256     }
257
258     @Override
259     public double calculateSalary()
260     {
261         return getBasicSalary() + (lecturesHandled * paymentPerLecture);
262     }
263 }

```

 Submitted

Qusestion-2



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QUESTIONS

Q1
Smart University Payroll Management System
25 marks

Q2
Smart Vehicle Insurance Claim System
25 marks

2/2 answered

Q2 Smart Vehicle Insurance Claim System

25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. Requirements Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks) Problem Statement An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. Requirements Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

Claim = insuredAmount × 0.80

Bike

Claim = insuredAmount × 0.60

Truck

Claim = insuredAmount × 0.90



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Your Code

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2
3 class Main
4 {
5     public static String readInput(Scanner sc)
6     {
7         String s="";
8
9         while(sc.hasNextLine())
10        {
11            s=sc.nextLine().trim();
12
13            if(!s.isEmpty())
14                return s;
15        }
16
17        return s;
18    }
19
20    public static void main(String args[])
21    {
22        Scanner sc=new Scanner(System.in);
23
24        int n=Integer.parseInt(readInput(sc));
25
26        Vehicle vehicle[]=new Vehicle[n];
27    }
```

Input

3

Car

TN10AB1234

Output

```
Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0
```

```
169
170     @Override
171     public double calculateClaimAmount()
172     {
173         return getInsuredAmount() * 0.60;
174     }
175 }
176
177 class Truck extends Vehicle
178 {
179     public Truck()
180     {
181
182     }
183
184     public Truck(String vehicleNumber,String ownerName,double insuredA
185     {
186         super(vehicleNumber,ownerName,insuredAmount);
187     }
188
189     @Override
190     public double calculateClaimAmount()
191     {
192         return getInsuredAmount() * 0.90;
193     }
194 }
```

 Submitted